

DSS Server

Use this plugin to monitor and control Dolby DSS Cinema Servers from within your Q-SYS designs.

Note: This plugin communicates with the DSS server using the Dolby SMI (System Management Interface) SOAP/HTTP API on port 8080. No login credentials are required.

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Compatibility

This plugin was developed and tested with Q-SYS Designer Software version 9.13.1 LTS and Dolby DSS software version 4.9.5.2.

Note: The Audio (Mute, Volume) controls on the Control page require that the *Dolby Cinema Processor* property be set to *Yes*. The Input Select controls require that the *CAT745 Mediablock* property be set to *Yes*.

Configuration Overview

Connect the Plugin

1. Drag the DSS Server plugin into your Q-SYS schematic.
2. In the plugin Properties, configure the *CAT745 Mediablock* and *Dolby Cinema Processor* settings to match your installation.
3. Save the design to your Core and run (F5).
4. On the *Setup* page of the plugin, enter the IP Address of the DSS server.

The plugin connects automatically once a valid IP address is entered. A Connection Status of **OK** confirms a successful connection. The DSS API does not require user credentials.

Note: The DSS server's SMI API is accessible on TCP port 8080. Ensure this port is reachable from the Q-SYS Core to the DSS server on your network.

Playback Modes

The DSS server supports two playback modes:

- **Manual:** Content must be selected and started manually via the plugin or the DSS server's own interface.

- **Scheduled (Automatic):** The DSS server manages content selection and playback according to its programmed schedule. Load buttons are hidden in this mode to prevent accidental disruption of scheduled programming.

Stop-Stop Function

The Stop button implements a two-press "Stop-Stop" function that mirrors the behavior of the DSS server's own interface:

- **First press (while Playing or Paused):** Stops playback and places the server in the Stopped state.
- **Second press (while Stopped):** Ejects the current content and immediately re-selects it, depending on the current mode:
 - In **Scheduled mode:** Ejects the content, briefly switches to Manual, then returns to Scheduled mode, re-arming the automation sequence.
 - In **Manual mode with a CPL loaded:** Ejects and reloads the same CPL.
 - In **Manual mode with an SPL loaded:** Ejects and reloads the same SPL.

Polling Behavior

The plugin uses an adaptive polling strategy to balance responsiveness with network load:

- While playing with more than 30 seconds remaining on the current CPL: polls every 15 seconds (configurable long rate).
- While playing with 30 seconds or less remaining on the current CPL, or while stopped/paused: polls at the short rate (default 3 seconds, configurable in Properties).
- A free-running 1-second timer interpolates SPL and CPL position between polls for smooth progress bar and time display updates.
- Content lists (SPL/CPL) are refreshed at most once every 60 seconds.
- Active ingest transfers are polled every 5 seconds via a dedicated timer.
- Volume, Mute, and Input Select (when configured) are polled every 3 seconds.

Short Poll Rate

Set an integer value from 1 to 30 (default 3) for the number of seconds between poll cycles when the server is stopped, paused, or within 30 seconds of a CPL transition. Lower values provide faster response to state changes; higher values reduce API traffic.

CAT745 Mediablock

Set to *Yes* if the DSS system uses a Dolby CAT745 mediablock. When enabled, the Control page displays the Real-D mode button and the Input Select section (Server, HDMI 1, HDMI 2). Also enables the CAT745 Detected indicator LED, which lights green when the connected mediablock is confirmed to be a CAT745. When set to *No*, these controls are not shown. Default is *No*.

Dolby Cinema Processor

Set to *Yes* if the installation includes a Dolby Cinema audio processor. When enabled, the Control page displays the Audio section with Mute, Volume level, Volume fader, and Volume Up/Down buttons. Default is *No*.

Stopped LED Behavior

Controls when the Stopped LED and Stop button illuminate:

- **Stopped only** (default): The Stopped indicator lights only when the player transport state is explicitly STOPPED.
- **Not Playing or Paused**: The Stopped indicator lights whenever the player is neither Playing nor Paused, which includes READY, STOPPED, UNSELECTED, RECOVERING, LATE, and UNAVAILABLE states.

Debug Print

Controls data printed to the Q-SYS debug output window. Options:

- **None** (default): No debug output.
- **Tx/Rx**: Transmitted and received SOAP messages are shown.
- **Tx**: Transmitted SOAP requests only.
- **Rx**: Received SOAP responses only.
- **Function Calls**: Each internal function posts a message when called.
- **All**: All of the above.

Controls

Setup Page

IP Address

Text entry field. Enter the IP address of the DSS server. The plugin connects automatically when a valid address is entered.

Serial Number

Text field, read only. Displays the serial number of the connected mediablock (DECODER component), e.g. the CAT862 or CAT745 chassis serial.

Software

Text field, read only. Displays the DSS software package version (e.g. 4.9.5.2) reported by the PLAYER component.

Server Time

Text field, read only. Displays the current date and time reported by the DSS server in ISO 8601 format, along with the server's configured time zone.

Connection Status

Displays the current connection state of the plugin:

- **OK**: Plugin is connected and communicating with the DSS server.
- **INITIALIZING**: Plugin is starting up and establishing the initial connection.

- **FAULT:** HTTP error received from the server. Check the IP address and network connectivity.
- **MISSING:** No device is responding at the specified IP address.

Credits

Displays plugin authorship and copyright information. This plugin was designed by Steve Gutttag and created with Claude AI.

Playback Page

Playback Mode — Manual / Scheduled

State trigger buttons. Indicate and control the DSS server's playback mode. The active mode button is illuminated. Click *Manual* to switch the server to manual control; click *Scheduled* to return to automated scheduling.

Note: The Load SPL and Load CPL buttons are hidden when the server is in Scheduled mode to prevent accidental content changes during a programmed show.

SPL Name

Text field, read only. Displays the name of the currently loaded Show Playlist (SPL).

SPL Progress

Meter, read only. Displays the percentage of the current SPL that has played, from 0% to 100%.

SPL Position

Text field, read only. Displays the elapsed time of the current SPL in hh:mm:ss format. Updated every second by a free-running interpolation timer between server polls.

SPL Remaining

Text field, read only. Displays the remaining time of the current SPL as -hh:mm:ss.

SPL Duration

Text field, read only. Displays the total duration of the current SPL in hh:mm:ss format.

Player Status LEDs

LED indicators, read only. Illuminate to show the current transport state of the DSS server. States and their colors:

- **Playing** (green): Server is actively playing content.
- **Paused** (amber): Playback is paused.
- **Stopped** (red): Playback has stopped. See the *Stopped LED Behavior* property for configuration options.
- **Seeking** (yellow): Server is seeking within the timeline.
- **Ready** (orange): Server is loaded and ready to play.
- **Selecting** (yellow): Server is in the process of selecting content.

- **Busy** (yellow): Server is performing a background process (e.g., preparing or purging a suite/KDM validation).
- **Recovering** (yellow): Server is recovering from an error condition.
- **Unselected** (light grey): No content is currently selected.
- **Unavailable** (red): The player is unavailable.

CPL Name

Text field, read only. Displays the human-readable title of the Composition Playlist (CPL) currently active in the playback timeline. If the CPL is not found in the content list (e.g., a lamp cue or internal automation clip), a lookup is attempted via the server's CPL document and the title is displayed when received.

CPL UUID

Text field, read only. Displays the unique identifier (UUID) of the currently active CPL.

CPL Progress

Meter, read only. Displays the percentage of the current CPL that has played.

CPL Type

Text field, read only. Displays the content type of the current CPL (e.g., FEATURE, TRAILER, TRANSITIONAL, POLICY).

CPL Position

Text field, read only. Displays the elapsed time of the current CPL in hh:mm:ss format.

CPL Remaining

Text field, read only. Displays the remaining time of the current CPL as -hh:mm:ss.

CPL Duration

Text field, read only. Displays the total duration of the current CPL in hh:mm:ss format.

Playback Controls

Round transport control buttons. The Play, Pause, and Stop buttons illuminate with their respective status colors (green, amber, red) when their state is active.

- **Skip Back** (|<): Trigger button. Skips to the previous CPL in the SPL timeline.
- **Play** (>): Trigger button. Initiates or resumes playback. Illuminates green when playing.
- **Pause** (||): Trigger button. Pauses current playback. Illuminates amber when paused.
- **Stop** (■): Trigger button. Stops playback. Implements the Stop-Stop function (see Configuration Overview). Illuminates red when stopped.
- **Eject** (▲): Trigger button. Deselects the current content, placing the server in the Unselected state.
- **Skip Forward** (>>|): Trigger button. Skips to the next CPL in the SPL timeline.

Select SPL

Combo box. Lists all available Show Playlists on the DSS server, sorted alphabetically. Select an SPL to load. The list is refreshed automatically every 60 seconds. Hidden when the server is in Scheduled mode.

Load SPL

Trigger button. Loads the SPL selected in the *Select SPL* combo box. Only available in Manual mode; hidden in Scheduled mode.

Select CPL / Clip

Combo box. Lists all available Composition Playlists (CPLs/Clips) on the DSS server, sorted alphabetically. Select a CPL to load directly. The list is refreshed every 60 seconds.

Load CPL

Trigger button. Loads the CPL selected in the *Select CPL / Clip* combo box. Only available in Manual mode; hidden in Scheduled mode.

Load Status

Text field, read only. Displays the status of the last load operation (e.g., "Loading SPL...", "Re-loading CPL..."). Clears automatically when the server confirms the requested content is loaded, or after 30 seconds.

Error

Text field, read only. Displays error messages from load operations, such as "Manual mode required" if a load is attempted while in Scheduled mode.

Content Page

Content List

List box. Displays all CPLs and Clips available on the DSS server, sorted alphabetically. Click an item to populate the Selected Content Info section below.

Name

Text field, read only. Displays the title of the selected content item.

UUID

Text field, read only. Displays the unique identifier of the selected content item.

Type

Text field, read only. Displays the CPL type of the selected content item (e.g., FEATURE, TRAILER, TRANSITIONAL).

Duration

Text field, read only. Displays the duration of the selected content item in hh:mm:ss format.

Encrypted

LED indicator, read only. Illuminates red when the selected content requires a valid KDM for playback.

Playable

LED indicator, read only. Illuminates green when the selected content is available for playback (has a valid KDM if required, and is fully ingested).

Ingest Page

Ingest Active

LED indicator, read only. Illuminates green when one or more content transfers are actively in progress. Dark red when idle. Polled every 5 seconds when a transfer is active.

Ingest Progress

Meter, read only. Displays the percentage complete of the highest-progress active transfer task.

Control Page

Note: The controls available on this page depend on the *CAT745 Mediablock* and *Dolby Cinema Processor* property settings.

Playback Mode (2D/3D) Buttons

Mutually exclusive StateTrigger buttons. Indicate and control the current projection mode. The active mode illuminates blue. Options:

- **2D:** Standard 2D projection.
- **3D:** Generic 3D projection.
- **Dolby 3D:** Dolby 3D projection system.
- **Real-D:** Real-D 3D projection system. Visible only when *CAT745 Mediablock* is set to Yes.

Mute

Toggle button. Illuminates red when the Dolby Cinema Processor audio output is muted. Click to toggle mute on or off. Visible only when *Dolby Cinema Processor* is set to Yes.

Volume Level

Text field, read only. Displays the current audio volume level (range 0.0 to 10.0). Updated every 3 seconds. Visible only when *Dolby Cinema Processor* is set to Yes.

Volume Up / Volume Down

Trigger buttons with arrow icons. Each press adjusts the volume by 0.1 in the respective direction. Visible only when *Dolby Cinema Processor* is set to Yes.

Volume Fader

Fader control, range 0.0 to 10.0. Drag to set the audio volume of the Dolby Cinema Processor. Updates the server immediately on release. Visible only when *Dolby Cinema Processor* is set to Yes.

Input Select Buttons

Mutually exclusive StateTrigger buttons. Visible only when *CAT745 Mediablock* is set to Yes. Indicate and control the active video/audio input source of the CAT745 mediablock. The active input illuminates orange. Options:

- **Server:** DCP content from the DSS server (default).
- **HDMI 1:** External HDMI input 1.
- **HDMI 2:** External HDMI input 2.

Input state is polled every 3 seconds. If a non-CAT745 mediablock is detected at runtime, these buttons are disabled.

CAT745 Detected

LED indicator, read only. Illuminates green when the DSS server's DECODER component is confirmed to be a CAT745 mediablock. Visible only when *CAT745 Mediablock* is set to Yes. If a different mediablock model is detected, the Input Select and Real-D buttons are disabled.

Control Pins

The following table lists the control pins available for this plugin when used in a Q-SYS schematic.

Pin Name	Value	String	Position	Direction
Setup				
IP Address	(text)			Input / Output
Serial Number	(text)			Output
Software	(text)			Output
Server Time	(text)			Output
Status	0 1 2 3 4 5	OK (Green) Compromised (Orange) Fault (Red) Not Present (Gray) Missing (Red) Initializing (Blue)	-	Output
Playback				
Manual Mode	0 1	Inactive Active	0 1	Input / Output
Scheduled Mode	0 1	Inactive Active	0 1	Input / Output

Pin Name	Value	String	Position	Direction
SPL Name	(text)			Output
SPL Progress	0.0 – 100.0 (%)			Output
SPL Position	(hh:mm:ss)			Output
SPL Remaining	(-hh:mm:ss)			Output
SPL Duration	(hh:mm:ss)			Output
SPL Position (sec)	(float, seconds)			Output
SPL Duration (sec)	(float, seconds)			Output
SPL Remaining (sec)	(float, seconds)			Output
Playing LED	0 / 1	Off / On	-	Output
Paused LED	0 / 1	Off / On	-	Output
Stopped LED	0 / 1	Off / On	-	Output
Ready / Selecting / Seeking / Busy / Recovering / Unselected / Unavailable LEDs	0 / 1	Off / On	-	Output
CPL Name	(text)			Output
CPL UUID	(text)			Output
CPL Progress	0.0 – 100.0 (%)			Output
CPL Type	(text)			Output
CPL Position	(hh:mm:ss)			Output
CPL Remaining	(-hh:mm:ss)			Output
CPL Duration	(hh:mm:ss)			Output
CPL Position (sec)	(float, seconds)			Output
CPL Duration (sec)	(float, seconds)			Output
CPL Remaining (sec)	(float, seconds)			Output
SkipBack / Play / Pause / Stop / Eject / SkipForward	trigger	-	-	Input / Output
Load SPL Name	(text / combo box)			Input / Output
Load SPL	trigger	-	-	Input / Output

Pin Name	Value	String	Position	Direction
Load CPL Name	(text / combo box)			Input / Output
Load CPL	trigger	-	-	Input / Output
Load Status	(text)			Output
Load Error	(text)			Output
Ingest				
Ingest Active	0 / 1	Idle / Active	-	Output
Ingest Progress	0.0 – 100.0 (%)			Output
Control				
2D Mode / 3D Mode / Dolby 3D Mode / Real-D Mode	0 / 1	Inactive / Active	0 / 1	Input / Output
Mute	0 / 1	Unmuted / Muted	0 / 1	Input / Output
Volume Level	(text, 0.0 – 10.0)			Output
Volume Fader	(float, 0.0 – 10.0)			Input / Output
Volume Up / Volume Down	trigger	-	-	Input / Output
Input Server / Input HDMI 1 / Input HDMI 2	0 / 1	Inactive / Active	0 / 1	Input / Output
CAT745 Detected	0 / 1	Not detected / Detected	-	Output

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